Question	Answer	Marks	Partial Marks
4(c)	12.7 or 12.73 to 12.74	**3**	**B2** for 2.26 or 2.261 to 2.262…. soi or **M2** for $\left(20^2 \times 15 - \frac{4}{3} \times \pi \times 6^3\right) \div 20^2$ oe or for $15 - \left(\frac{4}{3} \times \pi \times 6^3 \div 20^2\right)$ oe or **M1** for $20^2 \times 15 - \frac{4}{3} \times \pi \times 6^3$ oe or $20^2 \times D = \frac{4}{3} \times \pi \times 6^3$ oe If 0 scored, **SC1** for answer 11[.0] or 10.97 to 10.98
5(a)	20	**2**	**M1** for $11x = 10(x + 2)$ oe

Question	Answer	Marks	Partial Marks
5(b)(i)	$\frac{95}{y}+\frac{147}{y+2}=12$	**M2**	**M1** for $\frac{95}{y}$ or $\frac{147}{y+2}$
	$95(y+2)+147y=12y(y+2)$ oe	**M1**	Allow correct or for clearing *their* equation with algebraic fractions in y and $y + 2$ Allow $95y+190+147y=12y^2+24y$ oe
	leading to $6y^2 – 109y – 95 = 0$	**A1**	With all brackets shown expanded and no errors or omissions
5(b)(ii)	$(6y + 5)(y – 19)$	**2**	**B1** for $(6y + a)(y + b)$ with $ab = –95$ or $a + 6b = –109$ or $(3y + a)(2y + b)$ with $ab = –95$ or $2a + 3b = –109$ or for partial factorisation $y(6y + 5) – 19(6y + 5)$ or $6y(y – 19) + 5(y – 19)$
5(b)(iii)	19	**1**	Correct or FT *their* positive answer from factors dep on B1 earned
6	11.9 or 11.91 to 11.92	**7**	**B5** for $t = 1.055$ or 1.0550... **M1** for $\tan w=\frac{their\ t}{5}$ oe OR **M1** for $(2t+3)^2=t^2+5^2$ oe seen isw **M2** for $3t^2+12t-16[=0]$ oe seen isw or **B1** for $4t^2+6t+6t+9$ **M1FT** for $\frac{-12\pm\sqrt{12^2-4(3)(-16)}}{2(3)}$ oe **M1** for $\tan w=\frac{their\ t}{5}$ oe
7(a)(i)	21.5 or 21.52...	**2**	**M1** for $\tan(\ldots)=\frac{2.8}{7.1}$ oe
7(a)(ii)	10.2 or 10.17 to 10.18	**3**	**M2** for $\left(\frac{2.8}{\tan 21}\right)^2+7.1^2$ oe or **M1** for $\frac{2.8}{PR}=\tan 21$ oe